



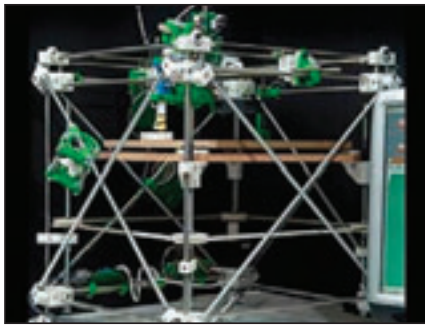
DARPA

3D printers are being used by Pentagon's DARPA to create insect-like wings for very small spy aircrafts – or 'robot-insects'

HP's history

In 1939, Dave Packard and Bill Hewlett established Hewlett-Packard (HP) in Packard's garage with an initial capital investment of US\$538

contact. Once the entire cross section of the build platform is covered, it lowers to allow the second layer to be formed in the same fashion. At the end of the process, the support parts – which are usually made of soluble materials – are dissolved in a chemical bath.



RepRap's FDM printers can be used to print themselves

Layer thickness is typically 0.005 to 0.013 of an inch and usually, the platform size is limited to a 10x10" or a 14x16" size. The process allows for complex parts to be made and is used by the famous RepRap project which specialises in making open-source 3D printers – enabling each and every person to make the printers themselves. What the Walkman did to the stereo music player, RepRap is quietly doing to the manufacturing industry. In fact, RepRap machines (like RepRap's 'Darwin' and MakerBot's 'CupCake CNC') are replicators – in the sense that you can use them to make the machine parts of the printer itself and put them together to make another one to sell to your friends. While they don't look all that great (apparently, only a bare skeletal frame), it's the ultimate in personalised production and has found its way into a lot of homes and academic institutions all over the world.

3D printing

While all of the above could be classified as 3D printers, the one that comes closest to our idea of 'printing' is the ink-jet technology, first created at MIT and currently one of the most popular commercial products in this category (eg. ZCorp, 3D Systems, SolidScape inc., etc.). Like the FDM system, a platform is set in a bin of powder material – depending on

the manufacturer, a whole lot of powders are available – starch based, ceramic, thermoplastic or even wax. An ink-jet head deposits a binder onto the powder to solidify the region of the layer as specified by the CAD. This fuses the powder together to create the bottom layer of the model. The platform is lowered and the process repeated until the model is finished. Extra powder serves to support the model and is blown or sucked off by vacuuming it. The excess powder is pneumatically removed and recycled for use in the next project. Then commences a process called 'infiltration'. Any gaps in the material are flaws and must be filled up. The ZCorp machine, for example, treats the model with a secondary resin material – usually a binder or epoxy – that is sprinkled or brushed on the faces of the model. This seeps into the miniscule nooks and crannies, driving the air out of them and sealing the model to give it a stronger and improved finish as it cures. Models can also be coloured, using technology similar to document printers by converting the PC's RGB space to a



ZCorporation's 3D printers are priced at less than 15,000 dollars

CMYK colour value. Unfortunately, the STL format doesn't support colour and you'll have to resort to others if you want your model painted. Besides, only the outer surfaces can be coloured as the machines are still not that advanced.

Laminated Object Manufacturing (LOM)

Helisys Inc. (now Cubic Technologies) came up with – and patented – a technique similar to lamination as another way of layering a model into shape. The machine contains spools of paper or plastic, laminated with heat-activated glue. The build process starts when a

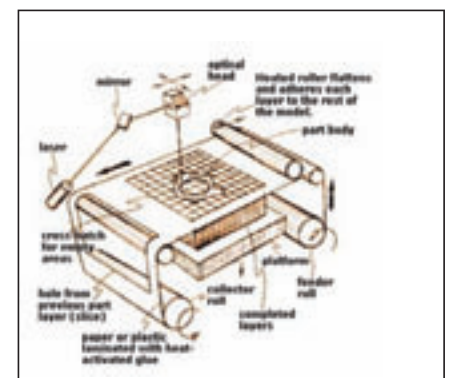
fresh layer of plastic is glued to the magnetic bed and a layer of glue spread over it. The build platform is usually constructed from paper and a double-sided foam tape but this varies from one printer to the other.

Since the entire space is not required to create the prototype, the machine spreads anti-glue over the excess material surrounding the object so that they can be removed later. Another layer of plastic is rolled out onto the first one. The glue binds them together. The excess material is cut usually using a blade or a laser (a process called 'cross-hatching'), as per the data of the design programme. The roller presses the two layers together and the process repeats until the object is created.

One of the most popular 3D printing machines is the FabLab project that provides equipment and support to enable students in universities to create prototypes from their designs. The size of the build platform (or 'bed-size') is typically limited to 160 x 200 mm so the models must be divided into smaller parts for greater efficiency.

Fore-casting

While the technology is still in a nascent stage, estimates predict a market of hundreds of millions of dollars for 3D printers by the end of 2013. Rapid Prototyping has quickly jumped ahead over the last few years and several new buzzwords and exciting new applications are already making the news. Rapid tooling, for example, is a process that uses CAD data to create tools using polymer-coated steel, plastic or a mixture of powdered metal to create tools. Sand-moulding



Laying it on thick